

PROJECT: BEE-1012

sub-system: EXP

Version: 1.0.0



Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH		Sheet title: 01_COVER PAGE.SchDocSheet 1 of 25	
Checked by: BAO BUI Q.		Document No: 1	Size: A3
Approved by: VU PHAM		Revision: V1.2.2	Date: 12/2/2025

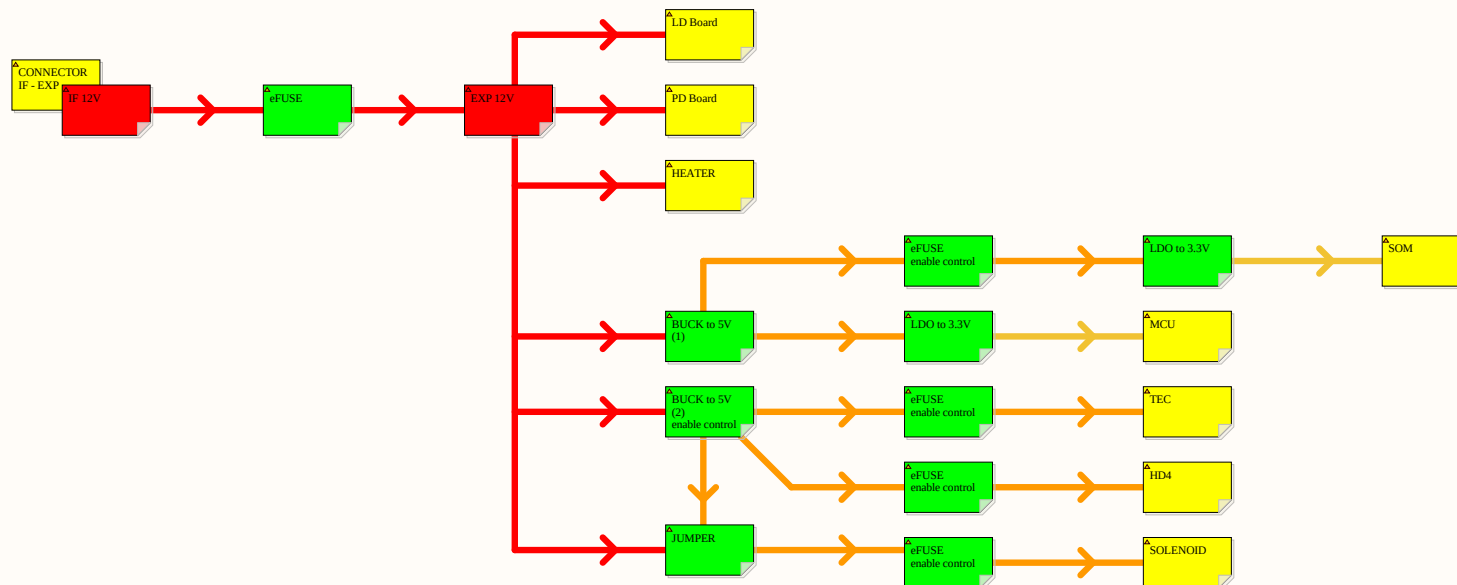
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D	<table><tr><td colspan="6">  </td></tr><tr><td colspan="3">Project: BEE-1012</td><td colspan="3">Sub-system: SATELLITE_EXP_CONTROLLER</td></tr><tr><td colspan="3">Design by: SANG HUYNH</td><td colspan="3">Sheet title: 02_BLOCK DIAGRAM.Sch Sheet 2 of 25</td></tr><tr><td colspan="3">Checked by: BAO BUI Q.</td><td colspan="2">Document No: 2</td><td>Size: A3</td></tr><tr><td colspan="3">Approved by: VU PHAM</td><td colspan="2">Revision: V1.2.2</td><td>Date: 12/2/2025</td></tr></table>												Project: BEE-1012			Sub-system: SATELLITE_EXP_CONTROLLER			Design by: SANG HUYNH			Sheet title: 02_BLOCK DIAGRAM.Sch Sheet 2 of 25			Checked by: BAO BUI Q.			Document No: 2		Size: A3	Approved by: VU PHAM			Revision: V1.2.2		Date: 12/2/2025
																																				
Project: BEE-1012			Sub-system: SATELLITE_EXP_CONTROLLER																																	
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Checked by: BAO BUI Q.			Document No: 2		Size: A3																															
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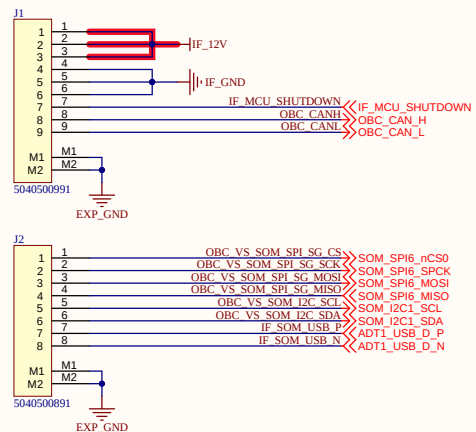


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Design by: SANG HUYNH	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 25	
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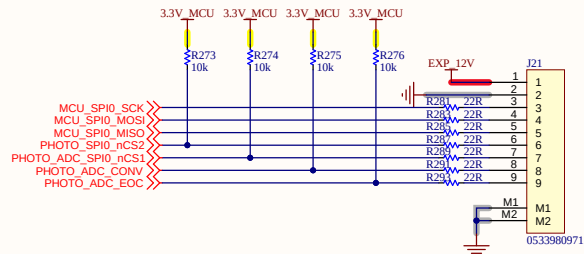
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Design by: SANG HUYNH	Sheet title: 04_POWER_TREE.SchDoc	Sheet 4 of 25
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form IF (OBC Board)

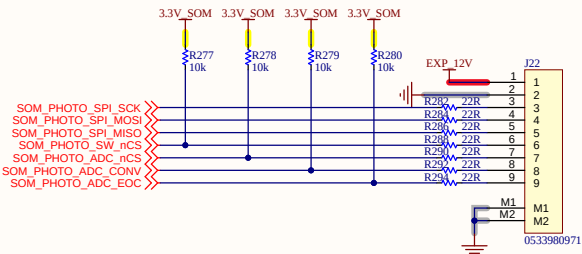


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Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_IN.SchDoc	Sheet 5 of 25
Checked by: BAO BUI Q.	Document No: 5	Size: A3
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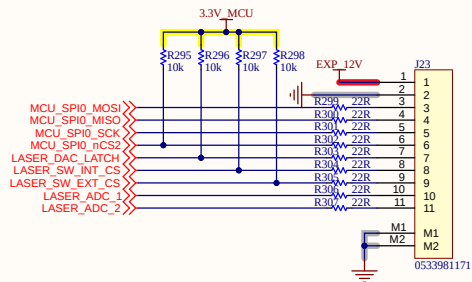
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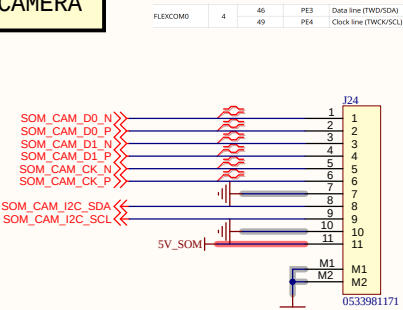
SOM to PHOTO



for LASER board



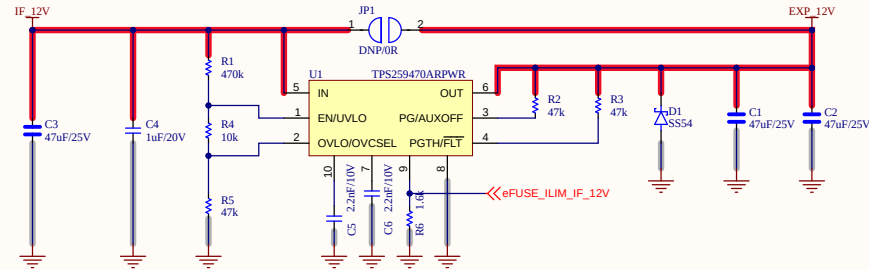
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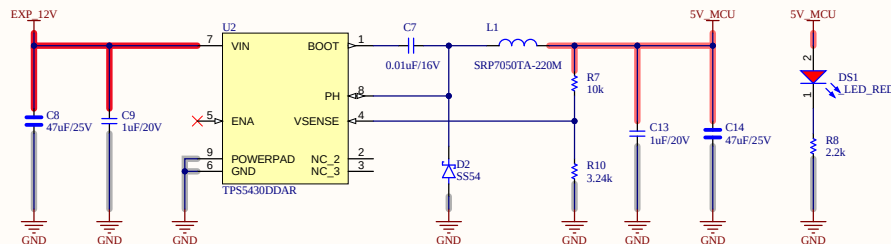
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Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_OUT.SchDoc	Sheet 6 of 25	
Checked by: BAO BUI Q.	Document No: 6	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025	

EFUSE 12V

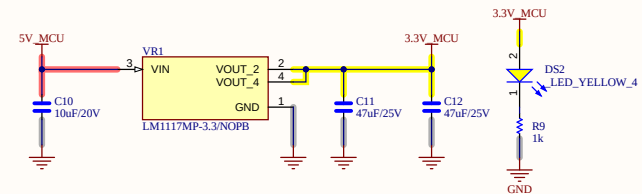
Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (47 + 470) / 47 = 13.2V$
 Slew Rate: $2000 / 2200 = 0.91V/ms$
 I-LIM: $3334 / 1600 = 2.08A$



BUCK for MCU, SOM

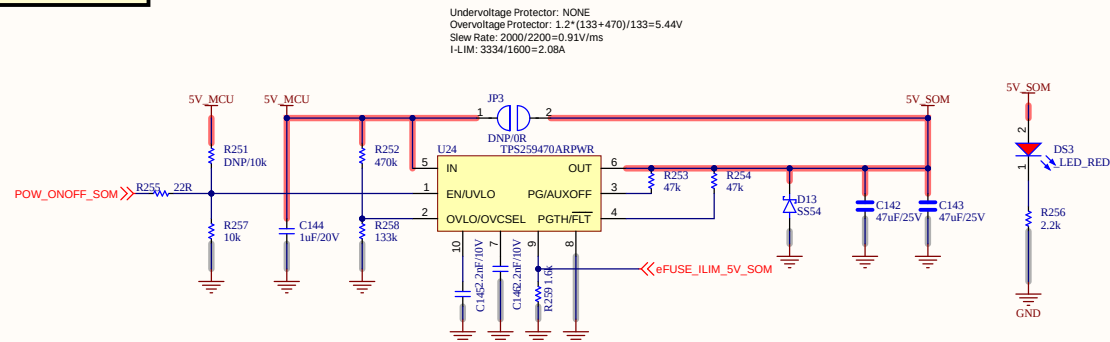


LDO for MCU

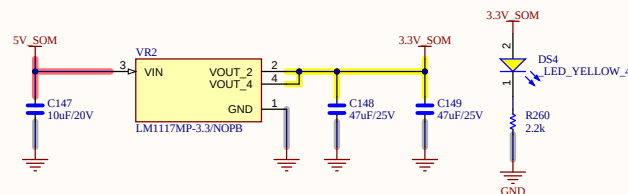


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EFUSE for SOM

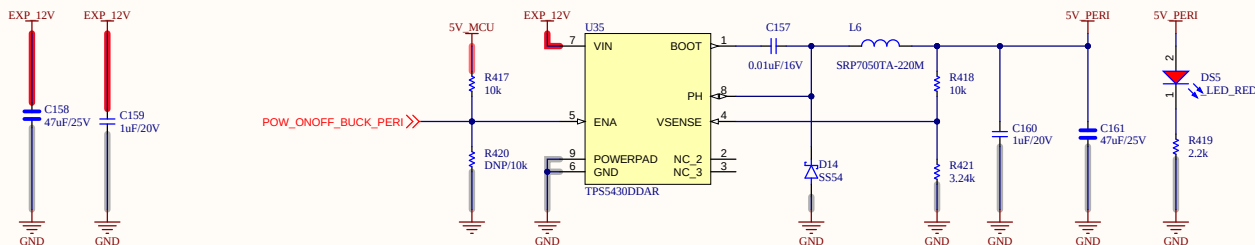


LD0 for SOM

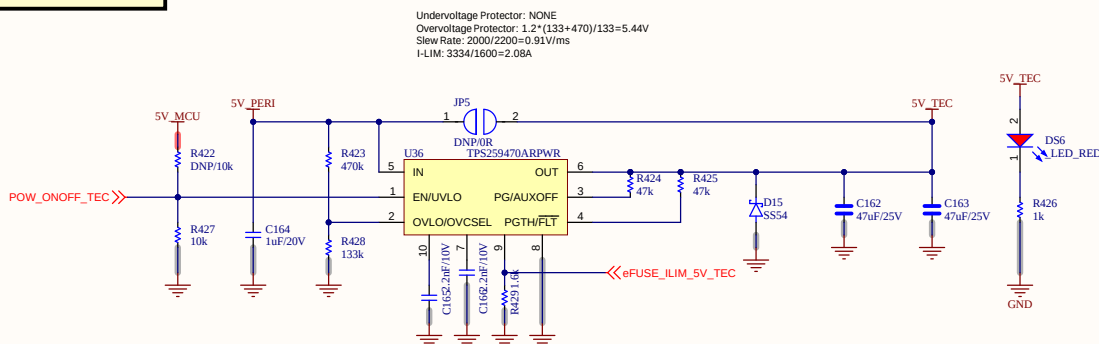


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BUCK for PERIPHERAL



EFUSE for TEC



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EFUSE for HD4

Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \cdot (133 + 470) / 133 = 5.44\text{V}$
 Slew Rate: $2000 / 2200 = 0.91\text{V/ms}$
 I-LIM: $3334 / 1600 = 2.08\text{A}$

The schematic diagram illustrates the EFUSE for HD4 circuit. It features a TPS259470ARPWPR (U37) undervoltage protector. The circuit is powered by 5V_MCU and 5V_PERI. The output is 5V_HD4, which is connected to a 5V_HD4 LED (DS7) through a 2.2k resistor (R434). The circuit also includes a 5V_HD4 EFUSE (eFUSE_ILIM_5V_HD4) and various passive components like resistors (R430, R435, R431, R432, R433, R434) and capacitors (C169, C170, C172, C173, C167, C168).

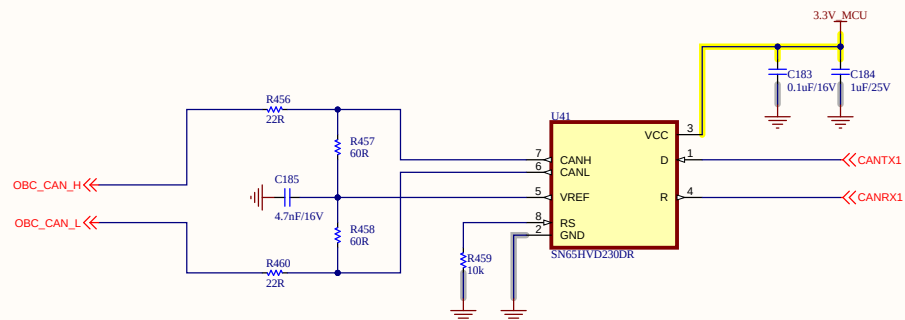
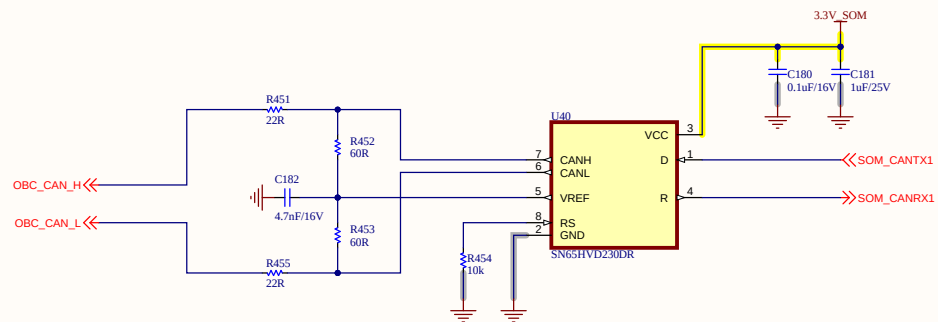
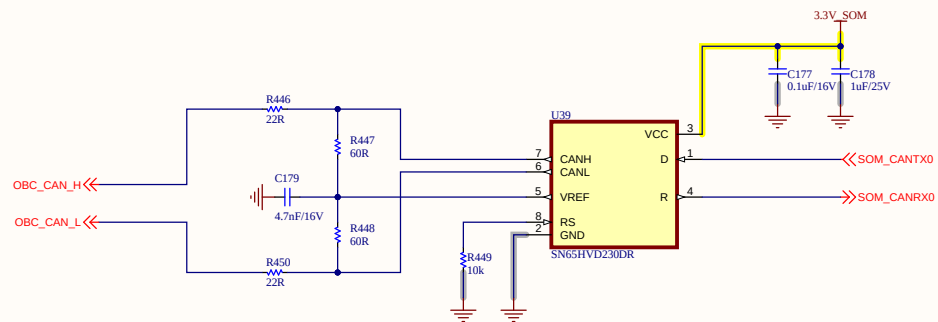
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Sub-system: SATELLITE_EXP_CONTROLLER

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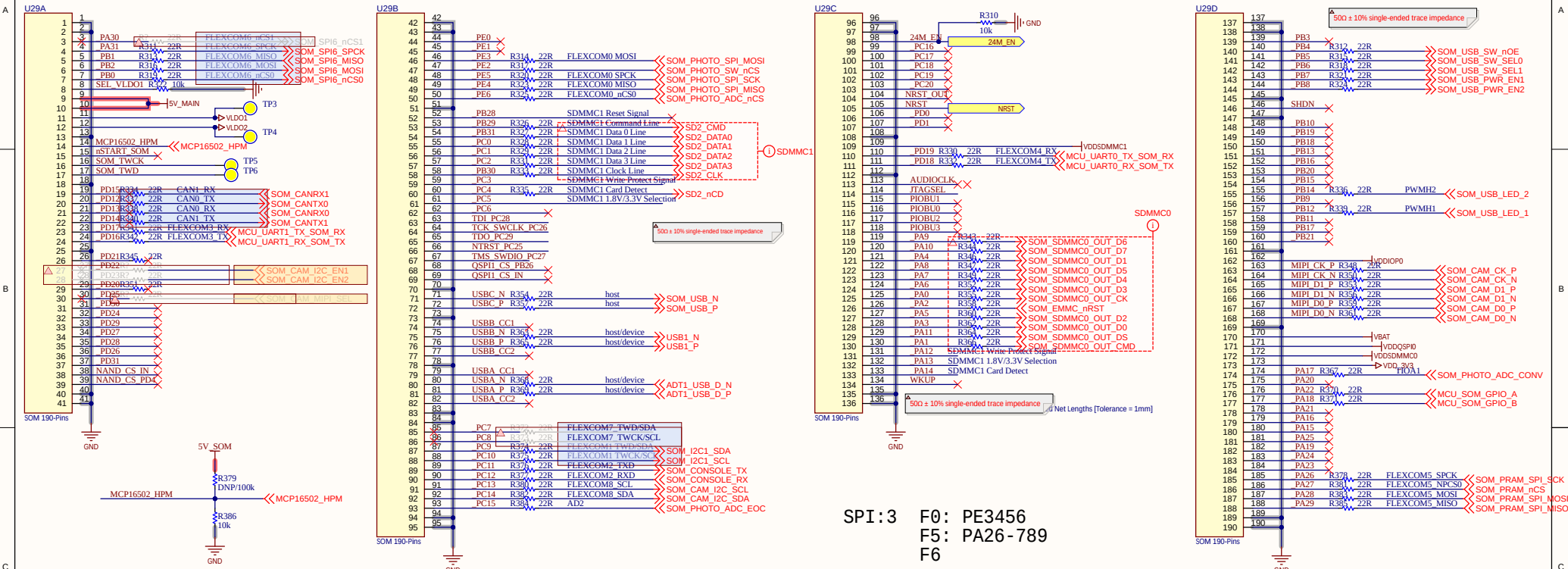
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SYSTEM ON MODULE PINOUT

MRAM dea SOM



SPI:3 F0: PE3456
F5: PA26-789
F6

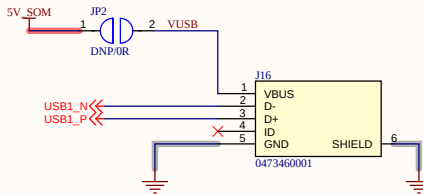
UART:3 F4:PD18 PD19
F3:PD16 PD17
F2:PC11 12

I2C:2 F1:PC9 10
F8:PC1314
F7:PC7 8

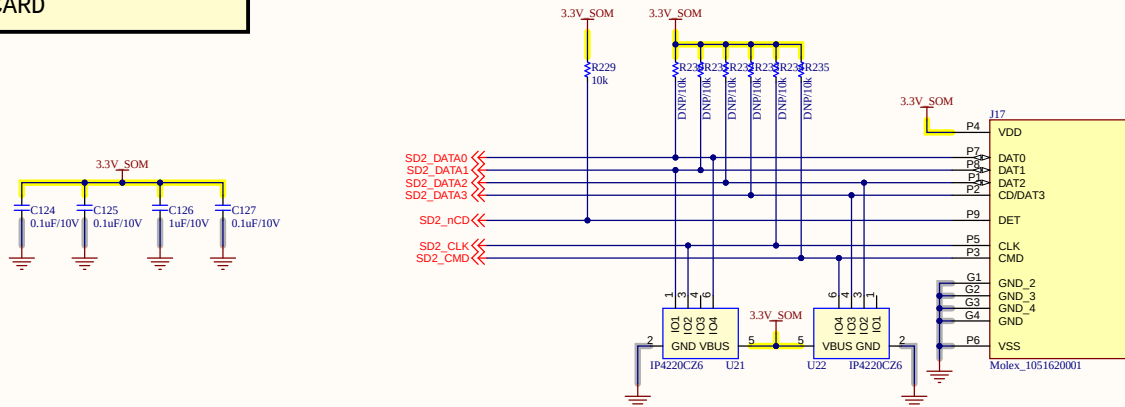


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USB

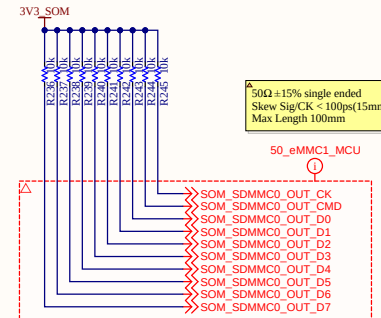
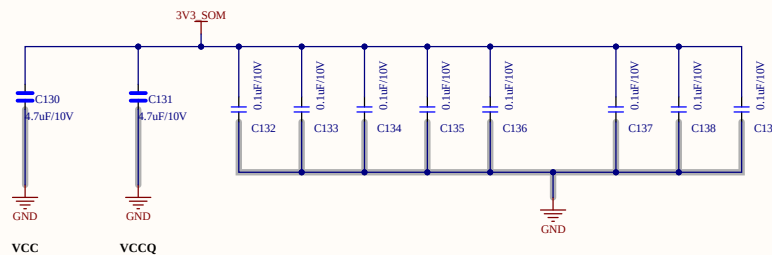
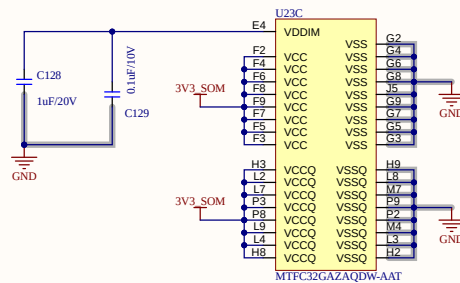
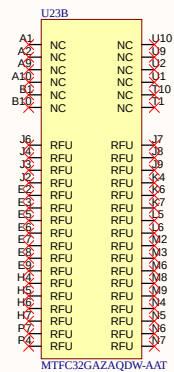
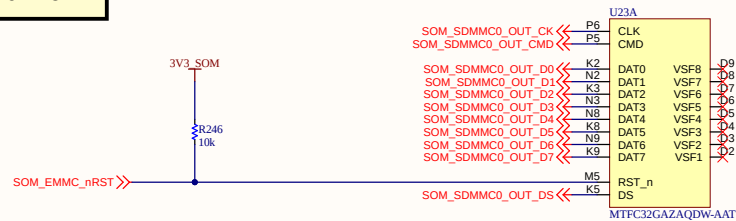


SD CARD



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eMMC - Buffer

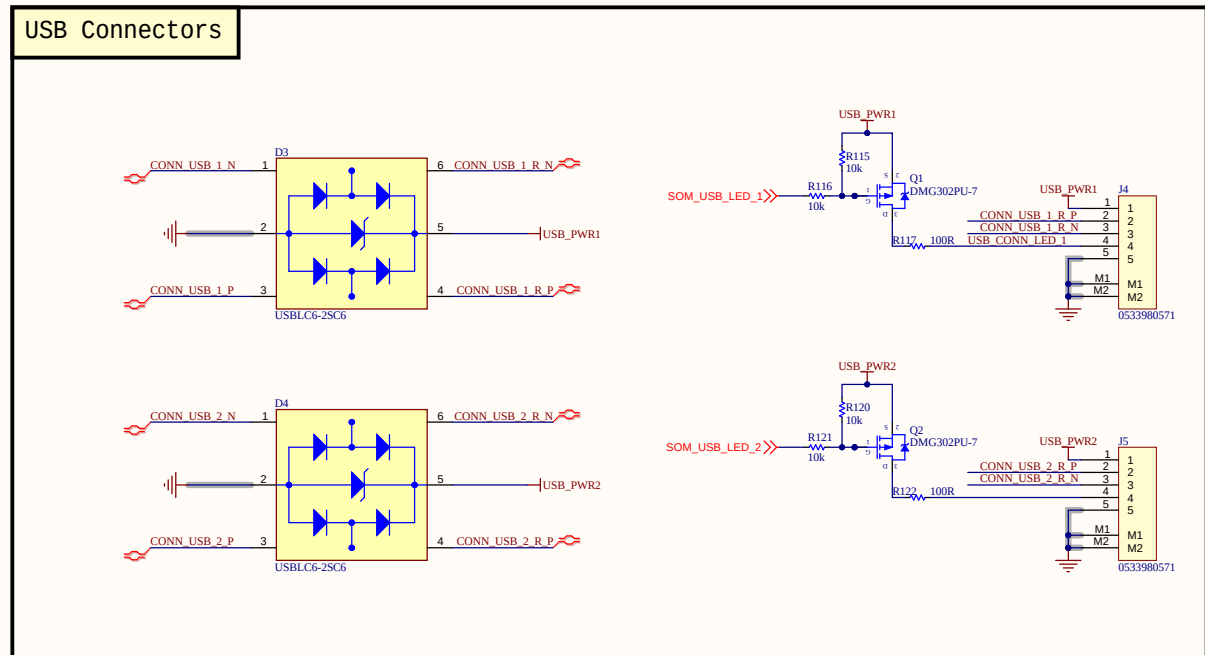
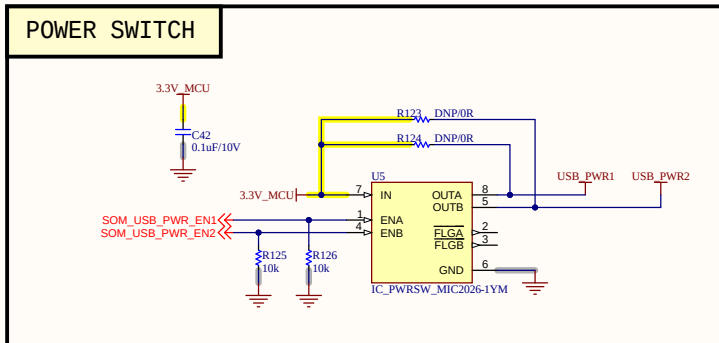
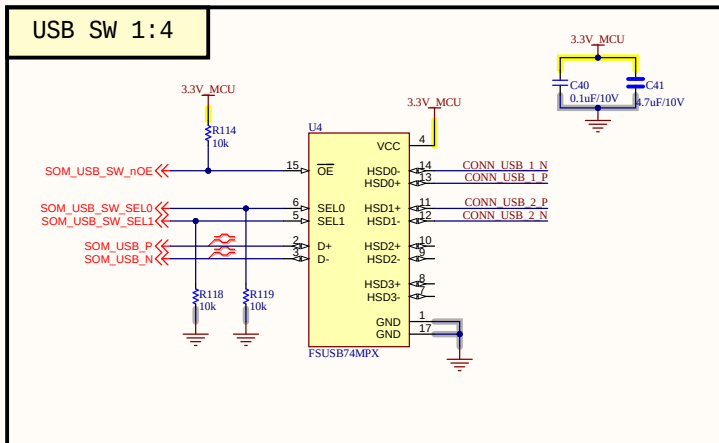


50Ω ±15% single ended
Skew Sig/CK < 100ps(15mm)
Max Length 100mm

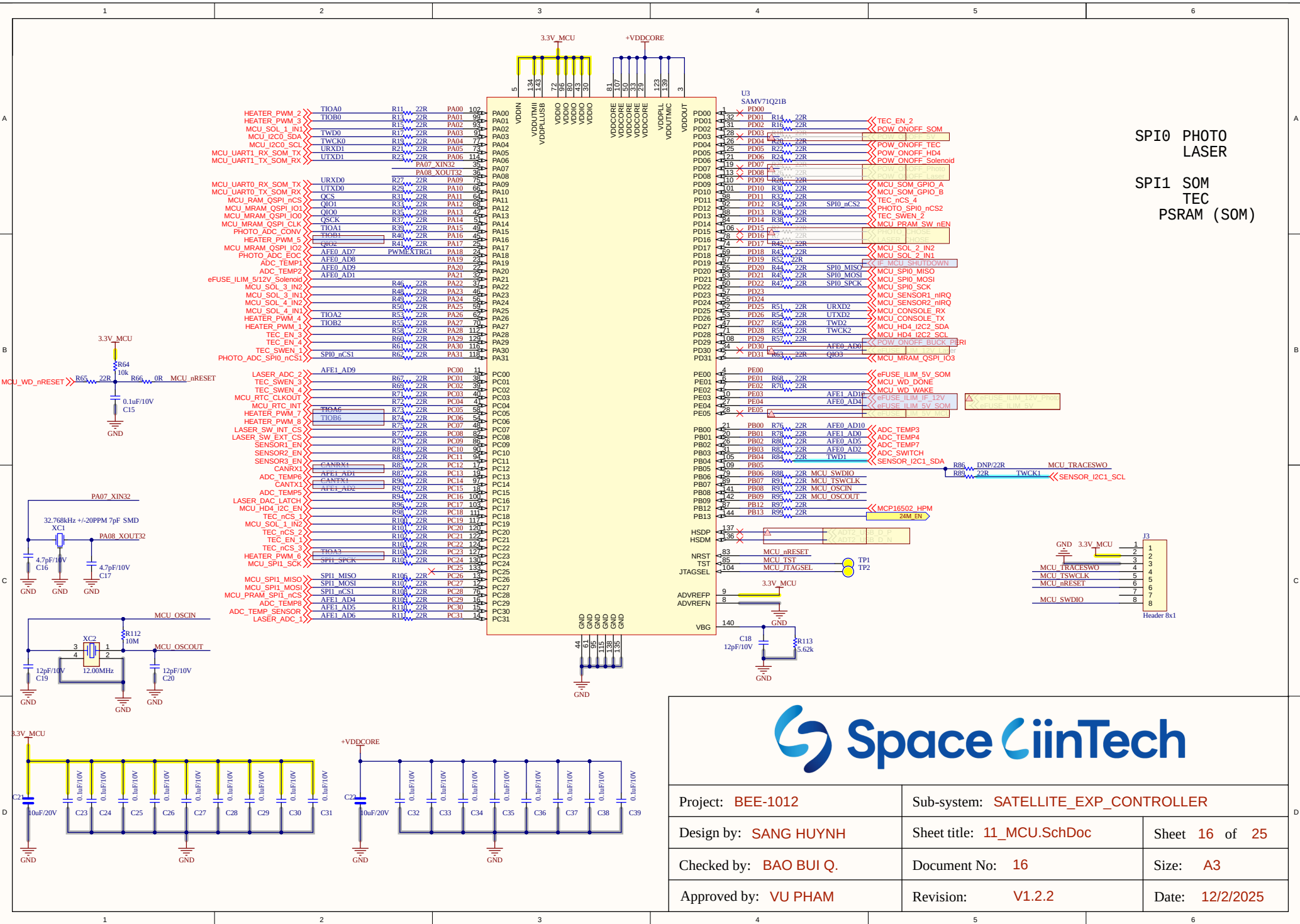
50_eMMC1_MCU



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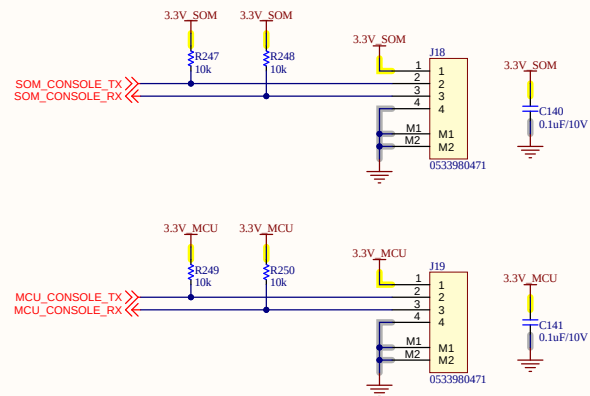
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Design by: SANG HUYNH	Sheet title: 10_USB.SchDoc	Sheet 15 of 25
Checked by: BAO BUI Q.	Document No: 15	Size: A3
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Design by: SANG HUYNH	Sheet title: 11_MCU.SchDoc	Sheet 16 of 25
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Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
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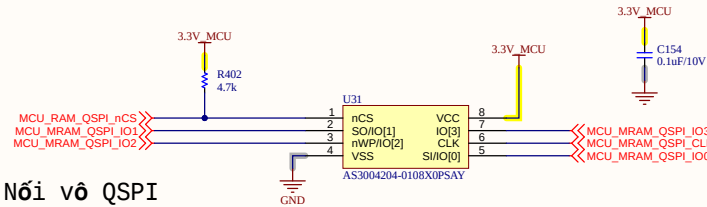
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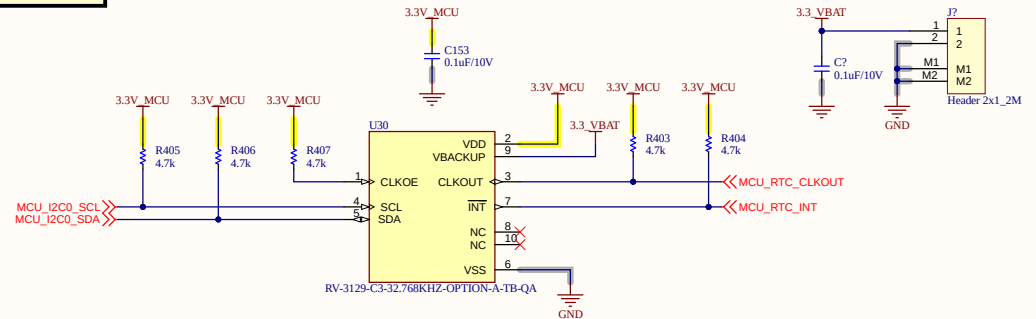
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MRAM 4Mbit

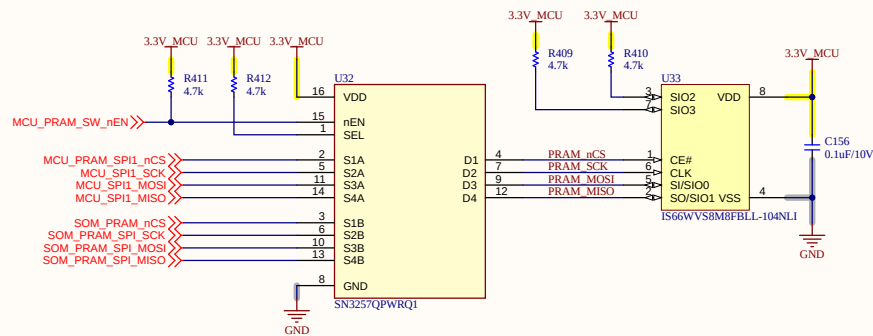
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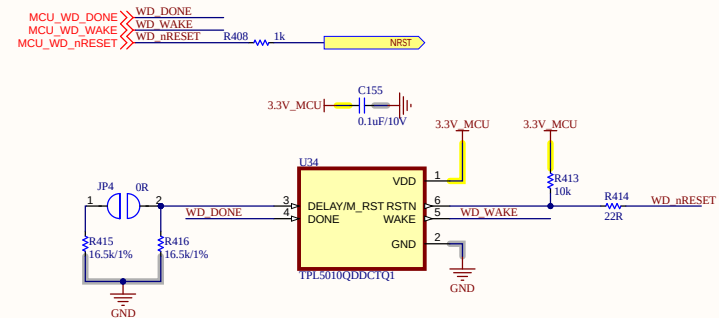
RTC



PRAM



WATCH DOG



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

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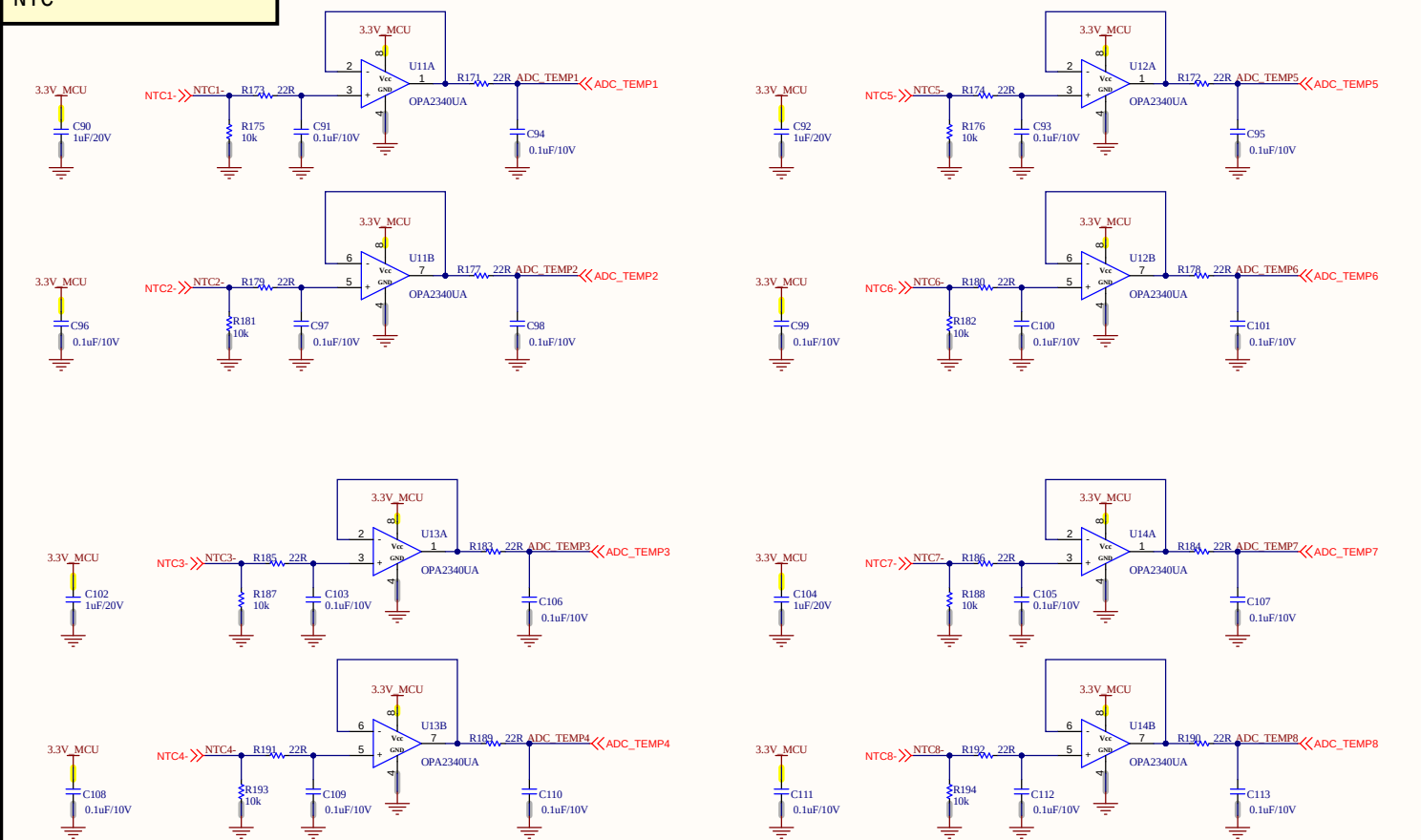
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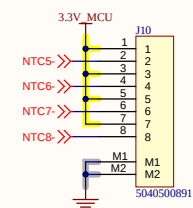
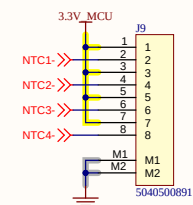
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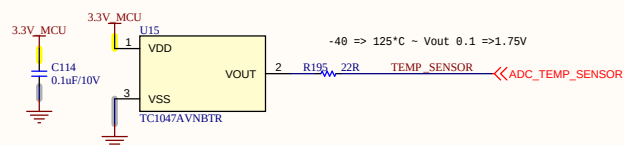
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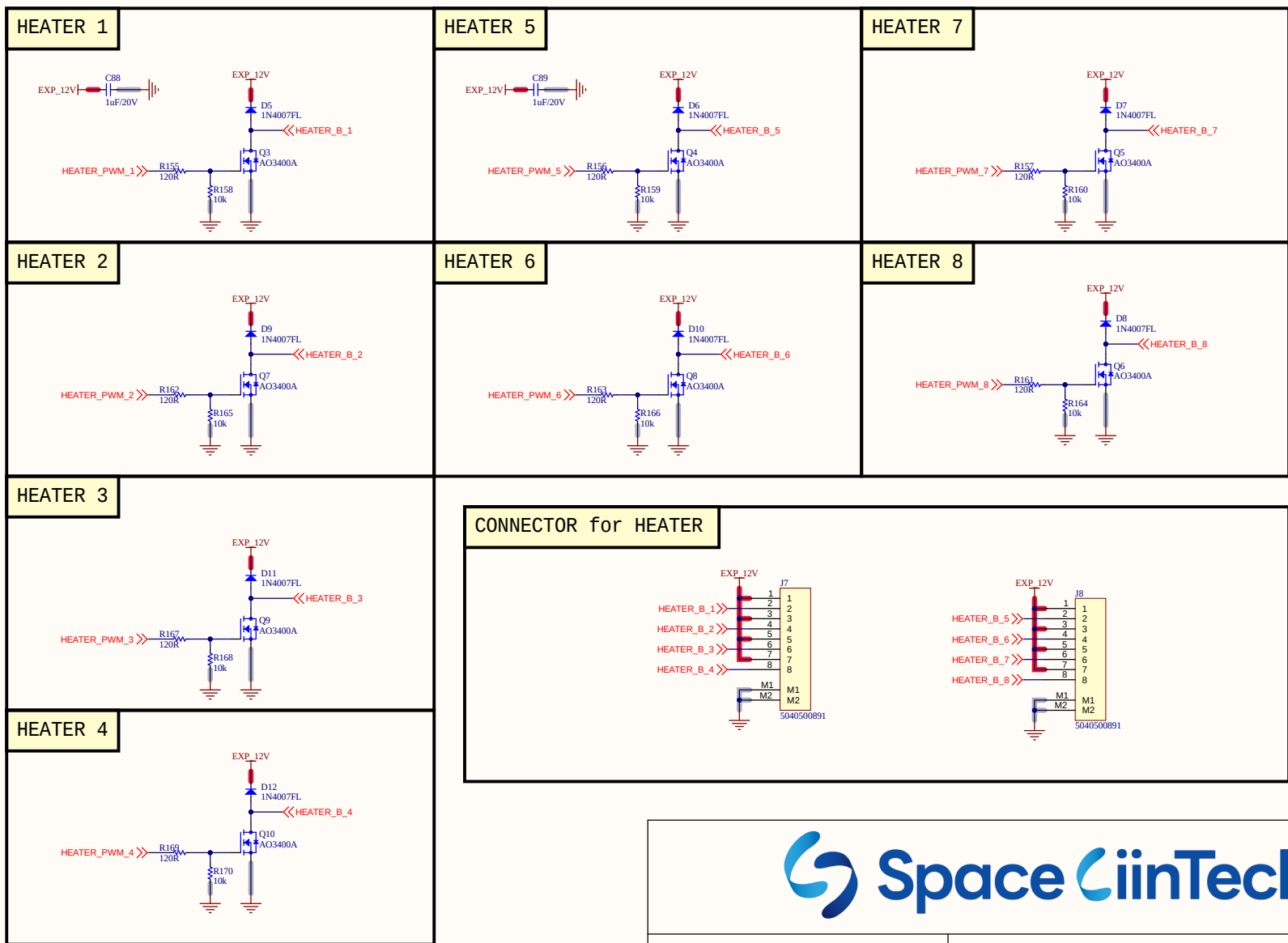
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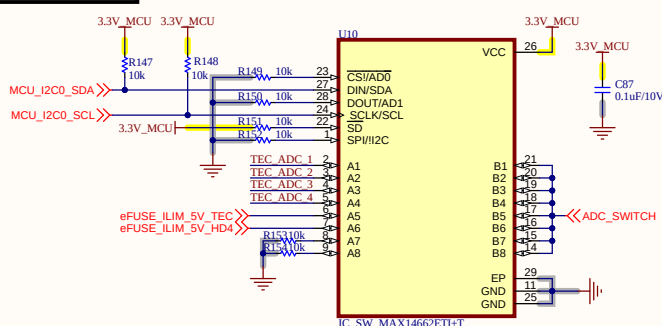
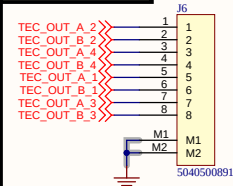
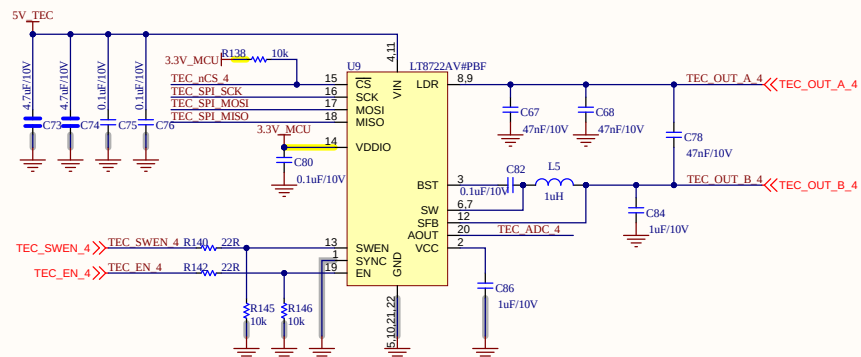
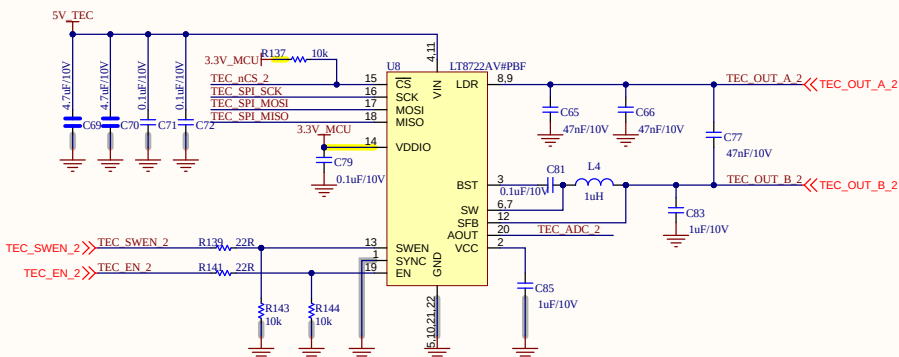
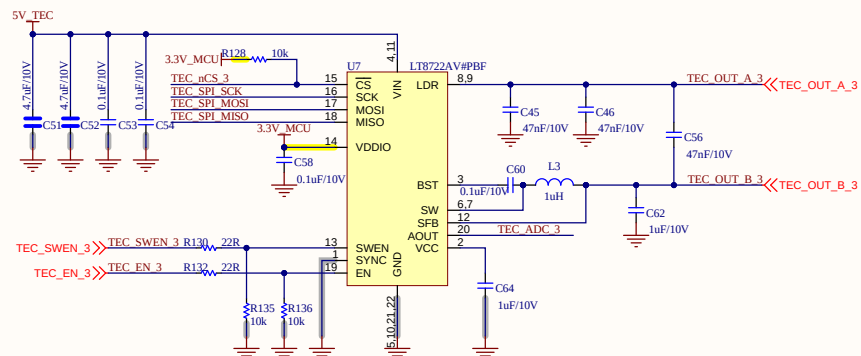
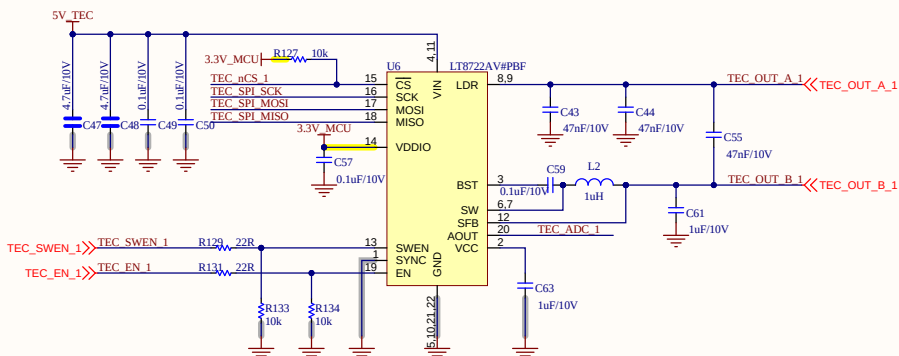
TEMPERATURE SENSOR



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Sub-system: SATELLITE EXP CONTROLLER

Design by: **SANG HUYNH**

Sheet title: 16_TEC.SchDoc

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Checked by: **BAO BUI Q.**

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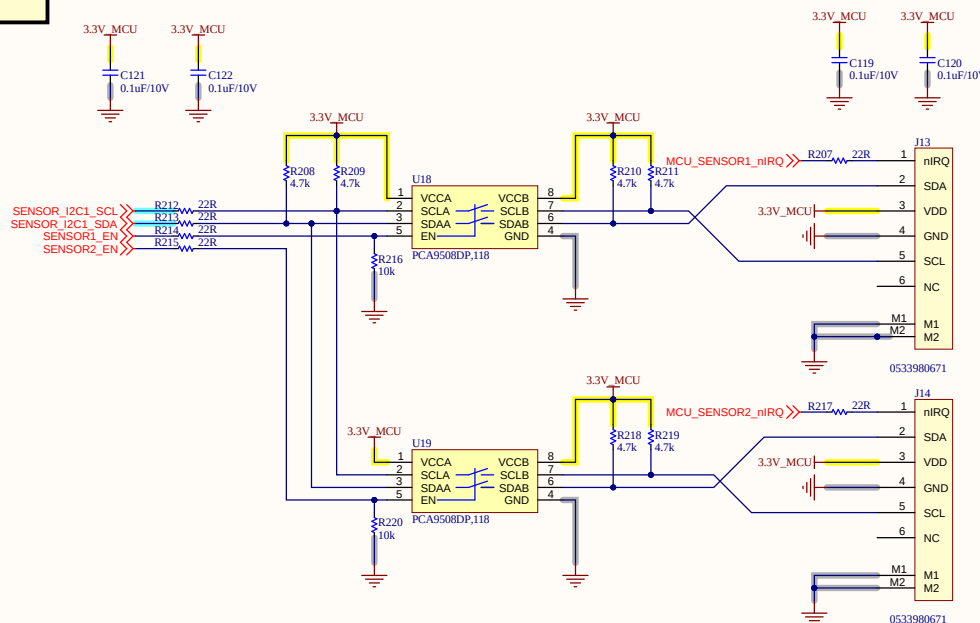
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Approved by: **VU PHAM**

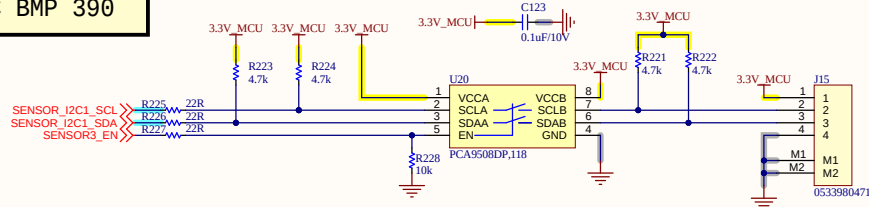
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I2C

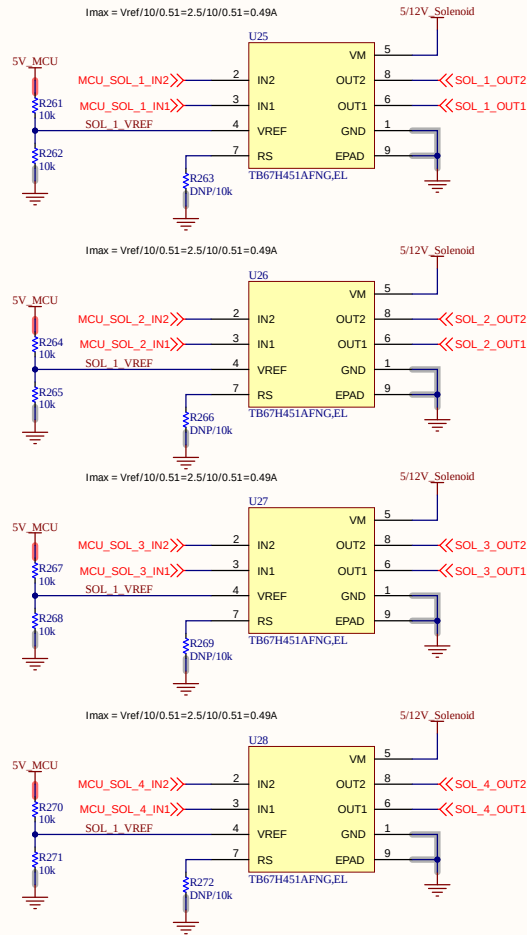


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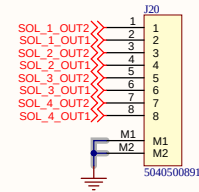


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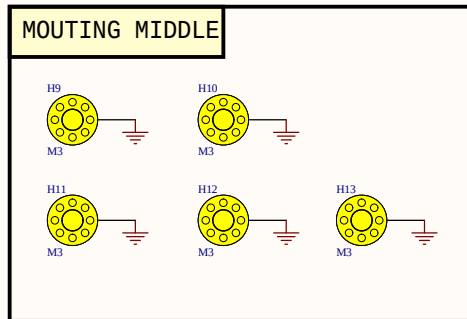
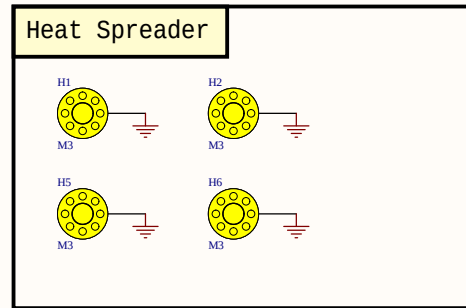
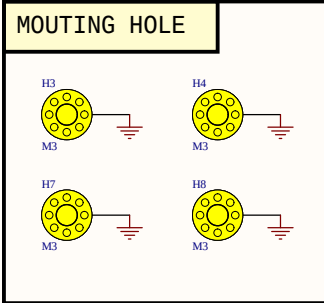
SOLENOID



CONNECTOR SOLENOID



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